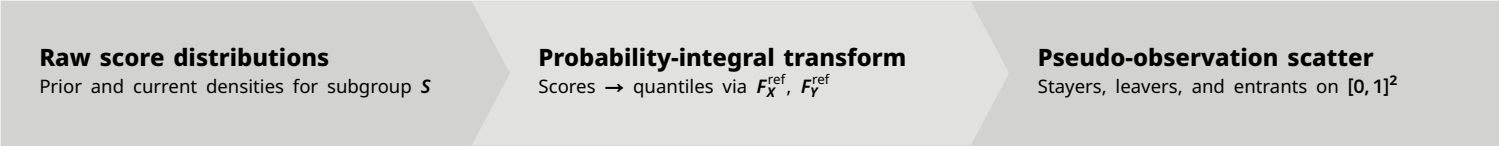


From Scores to Pseudo-Observations: The Marginal Transformation

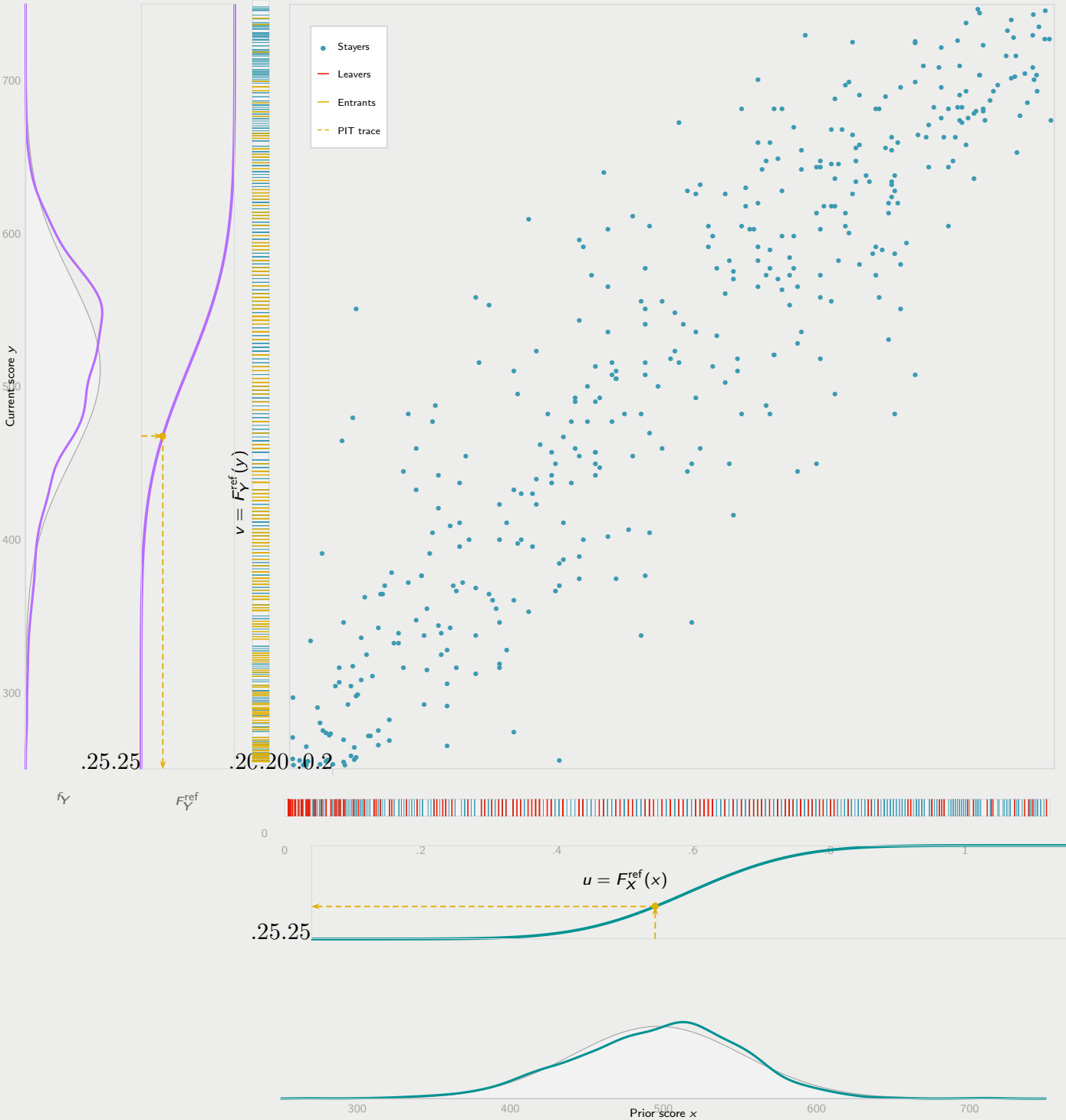
Stripping Metric Structure via the Probability-Integral Transform

Stayers, leavers, and entrants in copula-based longitudinal inference without longitudinal data



Pseudo-observation scatter: $u_i = F_X^{\text{ref}}(x_i)$ vs. $v_i = F_Y^{\text{ref}}(y_i)$

Subgroup S (2691 stayers, 147 leavers, 206 entrants; matching rate 93%)



The probability-integral transform. Suppose we observe prior scores $X = (x_1, \dots, x_{n_x})$ and current scores $Y = (y_1, \dots, y_{n_y})$ for a subgroup S (e.g., a TIMSS country or a NAEP state). The scores live on assessment-specific metric scales whose units carry no intrinsic cross-time meaning. To place both margins on a common, metric-free footing we apply fixed *reference* cumulative distribution functions (CDFs), computed over the full assessment population, to obtain *pseudo-observations* on $[0, 1]$:

$$u_i = F_X^{\text{ref}}(x_i), \quad v_j = F_Y^{\text{ref}}(y_j), \tag{M.1}$$

where F_X^{ref} and F_Y^{ref} are the prior-grade and current-grade population CDFs, respectively. This is the classical *probability-integral transform* (PIT): if F_X^{ref} is the true marginal CDF of X , then $U = F_X^{\text{ref}}(X)$ is uniformly distributed on $[0, 1]$.

Population versus subgroup. The reference CDFs are defined over the *full population*, not over S . Consequently, the marginal distribution of U within subgroup S is generally *not* uniform—it reflects S ’s position relative to the population. The marginal density plots in the outer margin strips illustrate this: the grey-shaded population density is symmetric and centered, while the colored subgroup density is shifted toward lower quantiles, reflecting a below-average subgroup. The population-level Uniform(0, 1) serves as the baseline against which subgroup departures are measured.

Stayers, leavers, and entrants. When prior and current scores come from the *same students* measured longitudinally, we observe linked pairs (x_i, y_i) —and hence pairs (u_i, v_i) —that carry information about how achievement at two time points co-varies within individuals. These *stayers* appear as scatter dots in the center panel and as rug marks on both axes. Formally, linked data are draws from a joint law,

$$(U_i, V_i) \stackrel{\text{i.i.d.}}{\sim} C(F_X^{\text{ref}}(x), F_Y^{\text{ref}}(y)), \tag{M.2}$$

where C is the copula encoding within-subject dependence (Sklar’s theorem). *Leavers*—students present in the prior round but not the current one—appear only as rug marks on the bottom axis. *Entrants*—students new to the current round—appear only as rug marks on the left axis. All three groups contribute to the marginal distributions, but only stayers contribute to the empirical copula.³

Non-linkage and cross-sectional designs. Cross-sectional assessment programs (NAEP, TIMSS, PIRLS) draw independent samples at each occasion. The analyst observes only the marginal distributions,

$$X_1, \dots, X_{n_x} \stackrel{\text{i.i.d.}}{\sim} F_X, \quad Y_1, \dots, Y_{n_y} \stackrel{\text{i.i.d.}}{\sim} F_Y, \tag{M.3}$$

with no within-subject pairing observed. In terms of the scatter plot, this corresponds to having only the rug plots (marginals) with no scatter dots (copula). The coupling—the copula C that would connect them—is entirely missing and must be supplied by an external model or assumption.⁴

What the marginals preserve and what they discard. The transform $x \mapsto u = F_X^{\text{ref}}(x)$ is monotone and (if F_X^{ref} is continuous) invertible, so no ordinal information is lost.¹ What is discarded is the *metric structure*: the difference $x_2 - x_1$ depends on the scale, but the rank ordering $u_1 < u_2$ is scale-invariant. This is precisely the property exploited by Sklar’s theorem, which factors any joint distribution $H(x, y)$ as

$$H(x, y) = C(F_X(x), F_Y(y)), \tag{M.4}$$

separating the marginal behavior (F_X, F_Y) from the dependence structure (C) . The CDF strips in the margin make this factorization visually explicit: the CDF curves are the “bridges” that carry raw scores into copula space, and the scatter in the interior reveals the dependence structure that remains after the marginals have been removed.

¹ The transform $x \mapsto u = F_X^{\text{ref}}(x)$ is invertible when F_X^{ref} is continuous (strictly increasing on its support). In discrete or mixed-continuous settings, the CDF is not strictly monotone and the generalized inverse $F^{-1}(u) = \inf\{x : F(x) \geq u\}$ is used. This distinction matters when working with score scales that have mass points (e.g., integer-valued test scores). In such cases, randomized ranks or mid-rank adjustments are sometimes applied to ensure a continuous marginal on $[0, 1]$; see Genest & Nešlehová (2007) and Rüschendorf (2009) for details on pseudo-observation construction under ties.

² The reference CDFs F_X^{ref} and F_Y^{ref} are computed over the *full assessment population*—the complete set of examinees across all subgroups in a given grade and year. They are *not* subgroup-specific. This ensures that all subgroups are mapped to the *same* reference scale, so that a quantile of $u = 0.30$ means “at the 30th percentile of the population” regardless of which subgroup the student belongs to. The subgroup’s marginal density of U then reflects its distributional position within the population.

³ Stayers are students observed in both the prior and current assessment rounds; their linked pairs (u_i, v_i) constitute draws from the copula $C(F_X^{\text{ref}}(x), F_Y^{\text{ref}}(y))$. Leavers appear only in the prior round (prior-only observations) and entrants appear only in the current round (current-only observations). All three groups contribute to the marginal distributions, but only stayers contribute to the empirical copula.

⁴ In cross-sectional assessment programs (NAEP, TIMSS, PIRLS), independent samples are drawn at each occasion and no student-level pairing exists. The analyst observes only the marginal distributions, so the copula C must be supplied externally—typically from canonical copula tables calibrated on longitudinal datasets in similar content areas and year-spans (see the Overview infographic, Panel B1). The non-linkage premium calculations in the companion appendix quantify the precision cost of this information loss.